

Life Orchestrator — Full Stack Report

Purpose: Authoritative reference of every technology layer in this project. Use this document to answer "what did you build with?" questions in interviews, applications, and portfolio presentations.

1. Application Overview

Field	Value
Project name	Life Orchestrator
Type	AI-powered life-management single-page application (SPA)
Status	Live on Google Cloud Run
Live URL	https://life-orchestrator-662200881058.us-central1.run.app
Repository	Ismailab1/Life-Orchestrator

One-sentence summary:

A local-first, conversational AI agent that orchestrates schedules, tracks relationship health, and manages a life inventory — all without a backend database.

2. Frontend

Layer	Technology	Version	Role
UI Framework	React	19.x	Component tree, virtual DOM, state
Language	TypeScript	~5.8	Type safety across all source files
Styling	Tailwind CSS	3.4.x	Utility-first atomic CSS
Markdown rendering	react-markdown	9.0.1	Renders AI-generated responses safely
Browser storage client	idb	8.x	IndexedDB wrapper (schema laid out, migration path ready)
Routing	react-router-dom	7.x	Client-side SPA routing
Build tool	Vite	6.x	ES module dev server, HMR, production bundler
PostCSS	autoprefixer + postcss	latest	Tailwind CSS pipeline
Typography plugin	@tailwindcss/typography	0.5.x	prose classes for AI markdown

Markup / markup-adjacent:

- HTML5 (index.html entry point)
- CSS3 (via Tailwind utility classes; no hand-written CSS files)

Notable frontend patterns:

- View state machine replaces React Router for navigation between app sections
- Inline editing pattern (click-to-edit, auto-save on blur) — no modal forms
- Optimistic UI — UI updates immediately, AI confirmation arrives asynchronously
- Streaming UX — AI responses render token-by-token via ReadableStream
- Multimodal input — text, Base64 images, and voice (microphone permission requested)

3. AI / Machine Learning Integration

Component	Technology	Notes
AI model	Google Gemini 2.5 Pro	Primary orchestration and reasoning model
SDK	@google/generative-ai 0.24.x	Official Google SDK for Gemini API
Interaction mode	Function calling (tool use)	AI calls pre-defined TypeScript executor functions
Context strategy	Prompt engineering as application logic	System prompt in constants.ts encodes all AI behavior
Temporal modes	Past / Today / Future	AI switches behavior: Reflection / Active / Planning
Streaming	generateContentStream	Token-by-token text delivery to UI

Abort support	AbortController	Cancels in-flight Gemini streams on user request or date change
Timeout protection	Custom withTimeout() wrapper	30s stream, 30s chunk, 60s orchestration timeouts
Memory system	Structured Memory[] in localStorage	AI reads persistent user preferences/facts across sessions
Security boundary	Executor function validation layer	AI cannot run arbitrary code; all actions route through typed schemas

Gemini tool functions implemented:

- save_day_plan — persists structured schedule
- update_relationship_status — modifies Kinship Ledger entries
- log_checkin — records relationship contact event
- save_memory — stores long-term user facts
- propose_orchestration — emits a structured schedule proposal card in chat
- get_life_context — reads current inventory/ledger state into AI context
- get_relationship_status — queries specific relationship health

4. Backend / Server

Component	Technology	Notes
Runtime	Node.js (Alpine)	Production container
HTTP framework	Express.js	Minimal web server
API proxy	http-proxy-middleware	Proxies /api/gemini/* → Google Generative Language API
Secret management	Cloud Run environment variables	GEMINI_API_KEY never bundled into client JS
Static file serving	express.static	Serves the Vite-built dist/ SPA
SPA fallback	app.get('*') catch-all	Returns index.html for client-side routes
CORS headers	Custom setHeaders middleware	Cross-Origin-Opener-Policy: same-origin-allow-popups for Google OAuth

5. Data Layer

Storage	Technology	What is stored
Primary persistence	Browser localStorage	Tasks, messages, ledger, memories, orchestration state
Future migration target	Browser IndexedDB (via idb)	Schema defined in services/db.ts; not yet activated
External calendar	Google Calendar API (GAPI + GSI)	Bi-directional event sync

No server-side database. All user data is local to the browser. This is an intentional privacy-by-design decision.

Storage quota management:

- Hard limit: ~5 MB per origin
- Real-time percentage bar with amber (80%) and red (90%) warnings
- Granular cleanup by date; surgical delete rather than full reset
- Image compression pipeline: 800×800 px, JPEG 70% quality ≈ 100 KB
- Base64 overhead accounted for (×1.33 multiplier)

6. External APIs & Cloud Services

Service	Provider	Purpose
Gemini AI API	Google Cloud LLM inference, function calling, streaming	
Google Calendar API (GAPI)	Google Cloud Read/write calendar events	
Google Identity Services (GSI)	Google Cloud OAuth 2.0 sign-in popup flow	
Cloud Run	Google Cloud Serverless container hosting	
Container Registry	Google Cloud Docker image storage	

7. Infrastructure & Deployment

Component	Technology	Notes
Containerization	Docker (multi-stage)	Stage 1: Node 22 Alpine build; Stage 2: lean production image
Container orchestration	Google Cloud Run	Serverless, auto-scaling, zero cold-start cost
Deployment automation	PowerShell script (advanced_deploy.ps1)	Local build → push → Cloud Run deploy
Reverse proxy / static serving	Express.js (built-in)	No nginx in final image
Ignored build artifacts	.gcloudignore	Excludes node_modules, .git, source files from Cloud submission

Docker build strategy:

Stage 1 (build) → Node 22 Alpine → npm ci → vite build → dist/
Stage 2 (serve) → Node 22 Alpine → copy dist/ + server/ → npm install --omit=dev → node index.js

8. Developer Tooling

Tool	Purpose
TypeScript	Compile-time type checking across all source
Vite	Dev server (HMR <50ms) + production bundler
Tailwind CSS CLI	CSS purge + JIT compilation
PostCSS + Autoprefixer	Cross-browser CSS compatibility
.env + Vite define	Build-time env variable injection (dev only)

9. Key Source Files Reference

File	Role
App.tsx	Root component; state machine, all executor functions, AI stream handler
constants.ts	System prompt (SYSTEM_INSTRUCTION), temporal mode instructions
types.ts	Full data model: Task, Person, Memory, ChatMessage, OrchestrationProposal
services/geminiService.ts	Gemini SDK wrapper; chat session, streaming, tool definitions
services/storageService.ts	localStorage read/write helpers, quota management
services/googleCalendarService.ts	GAPI/GSI OAuth + calendar event CRUD
services/imageService.ts	Image compression pipeline
services/db.ts	IndexedDB schema (future migration)
components/ChatInterface.tsx	Conversational UI, message stream rendering
components/KinshipLedger.tsx	Relationship tracking UI
components/CareerInventory.tsx	Task/inventory management UI
components/LandingPage.tsx	Marketing/onboarding page
server/index.js	Express proxy server (production)
Dockerfile	Multi-stage container definition
vite.config.ts	Build config, CORS headers, code splitting
tailwind.config.js	Content paths, typography plugin

10. Lines of Code Snapshot (approximate)

Area	Approx. LOC
App.tsx (root logic)	~1,200+
services/	~700+
components/ (9 files)	~1,500+
types.ts + constants.ts	~400+

server/index.js ~90
Total (TypeScript/TSX/JS) ~4,000+

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